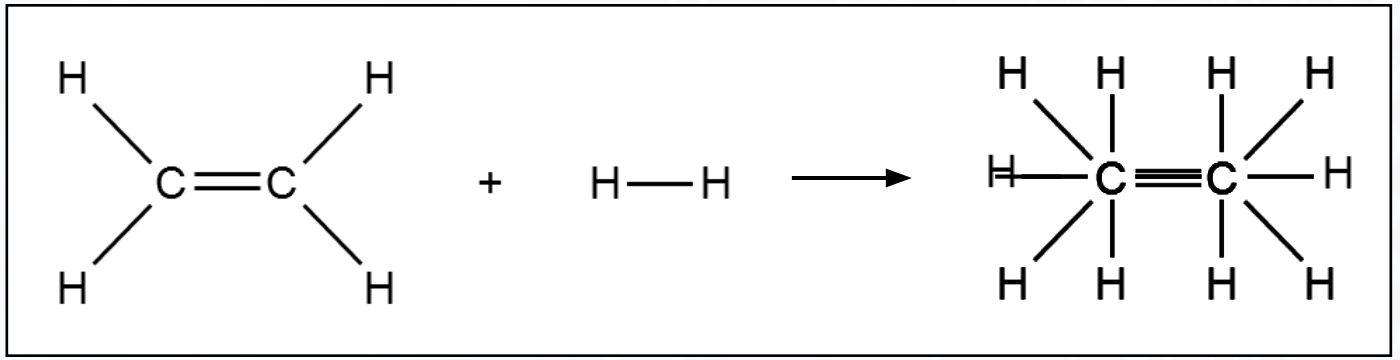
Mr Dimelow

3.1 Engineering materials

3.1.1.2 Polymers

Subject content	Additional information	Mathematical understanding
Thermoplastics: • ABS • acrylic • nylon • polycarbonate • polystyrene. Thermosetting polymers: • epoxy • polyester and melamine resins • polyurethanes • vulcanised rubber.	Students will not be expected to have practical experience of working with all of these polymers but exam questions could refer to any of the properties listed. The effects of heat on thermosets and thermoplastics.	

Polymers



In this presentation you will:

- explore the structure, characteristics and application of polymers

Introduction

Polymers, including the commonly known plastics, are an ever- present part of modern society.

Polymer materials have a huge range of applications ranging from ***disposable cups to body armour***.

The chemical structure of polymers makes them versatile materials with a wide range of properties.

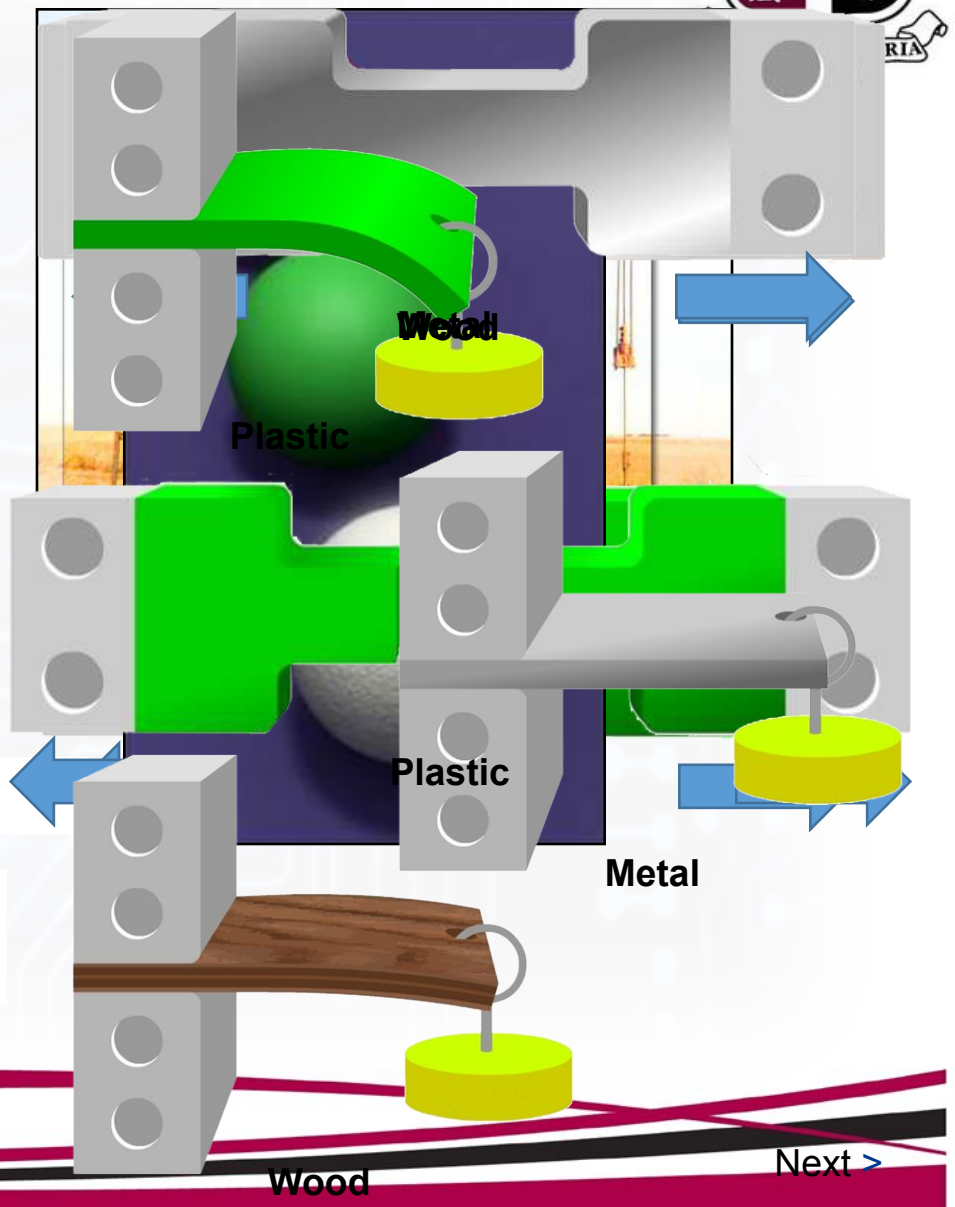


Next >

General Characteristics

Information about plastics:

- Produced from crude oil, coal, and natural gas
- Often referred to as polymers
- Available in a wide range of colours and surface finishes
- Resistant to rotting and corrosion
- Generally stronger than timber
- Generally weaker than metals
- Generally less stiff than either metal or timber



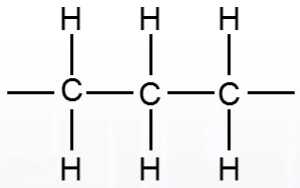
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Polymers

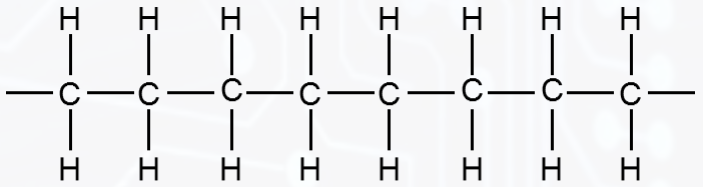
Polymers are very large organic molecules consisting of long chains of repeating subunits called ***mers***.

The length of a polymer chain can range from 77 to 750 mers.

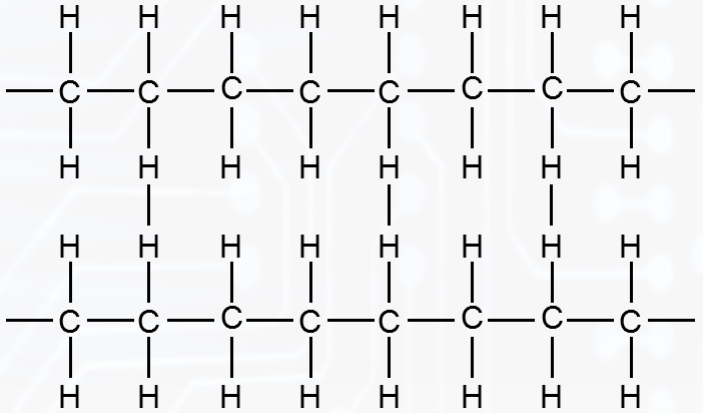
A polymer chain may be linked to other chains, forming a polymer molecule thousands of mers long.



Short polymer chain

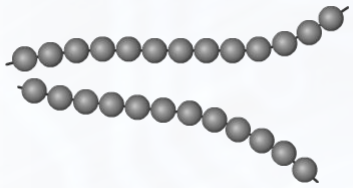


Long polymer chain



Long polymer chains linked together

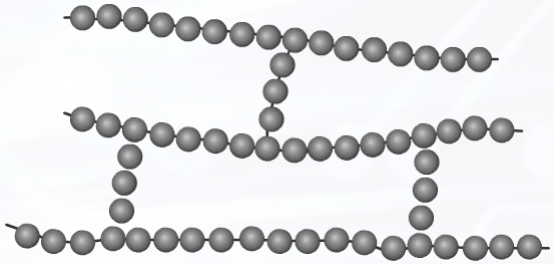
Polymer Structure



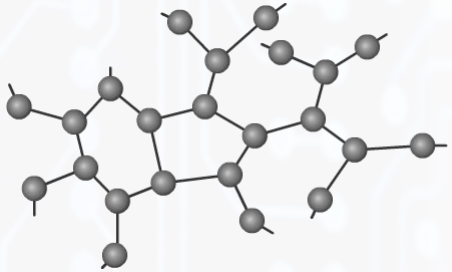
Linear



Branched



Cross linked



Network

Polymers have four different structures.

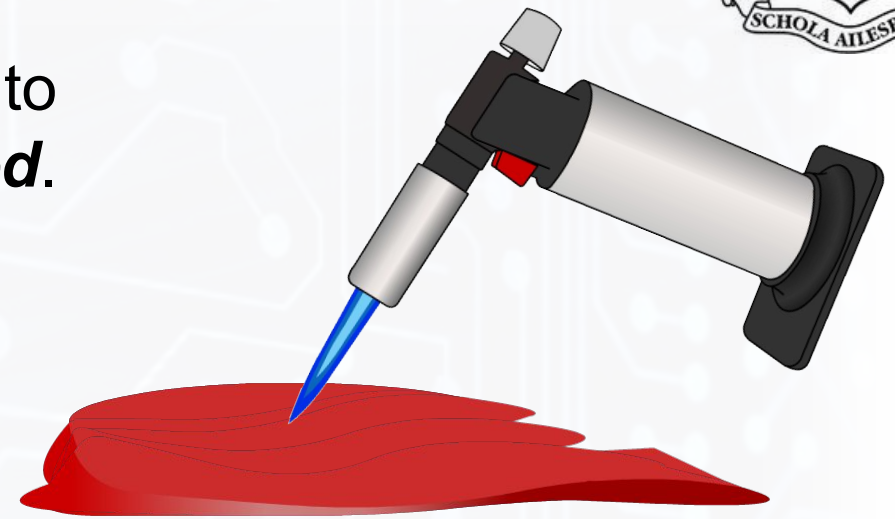
The type of structure influences the properties of the polymer.

Next >

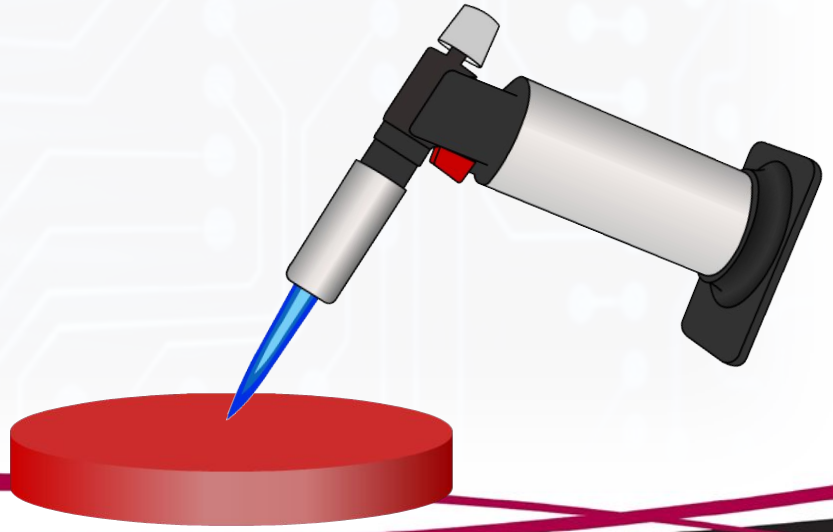
Classification of Polymers

The most common way to classify polymers is according to how they ***behave when heated***.

Thermoplastic polymers can be melted and reformed many times over.



Thermoset polymers can only be formed once, remaining solid when heated.

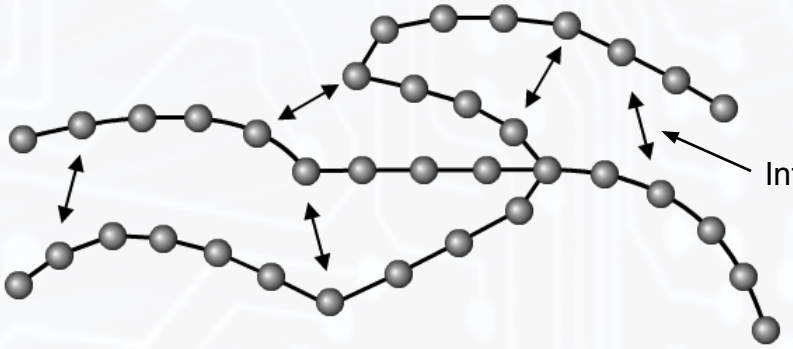


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Thermoplastic Polymers

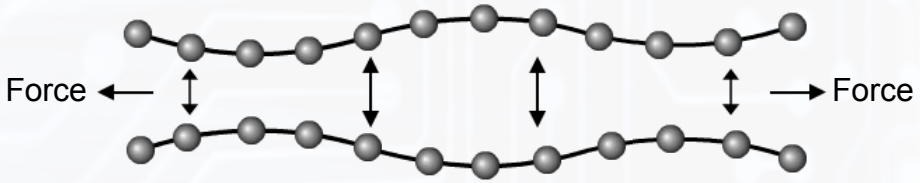
Thermoplastic Polymers

Thermoplastic polymers are formed from long, thin, chain molecules held together by intermolecular bonds.



These bonds weaken when heated, allowing ***thermoplastics*** to be *moulded easily*. When cooled, they re-harden.

There is no chemical change during this process; ***it can be repeated indefinitely***.

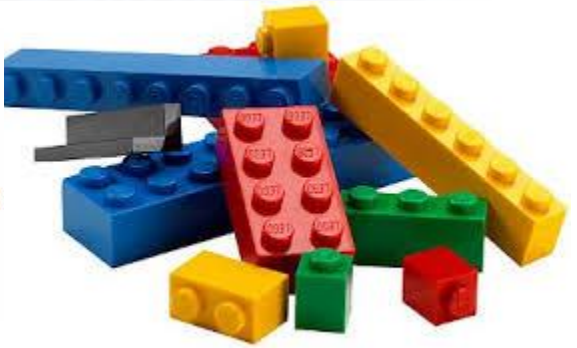


Next >

Acrylonitrile Butadiene Styrene (ABS)

ABS is an ideal material wherever **excellent surface quality, colourfastness** and ***shine*** are required.





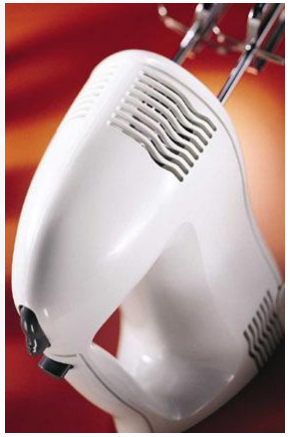
Properties

High impact strength and toughness, scratch resistance, light and durable, good appearance, high surface finish, resistant to chemical

Applications

Kitchen ware, cases for consumer durables (e.g. cameras), toys, safety helmets, car components, telephones, food processes/mixers





Next >

Acrylic

Acrylic is a useful, ***clear*** plastic that ***resembles glass***, but has properties that make it superior to glass in many ways. In the UK ***acrylic*** is often known by its brand name ***'Perspex'***.

As well as transparent acrylic can also be ***'translucent'*** or ***'opaque'***.

Properties

Stiff, hard, clear, very durable, 10x impact resistance of glass but scratches easily, excellent light transmission, fibre optic qualities, safe with food, good electrical insulator, colours well, easily machined, easily bent/shaped (when heated), polishes well





Applications

Light units, illuminated signs, aircraft canopies, windows, rear car lights/reflectors, furniture, machine guards





Next >

Nylons (Polyamide)

The name ***"nylons"*** refers to the group of plastics known as ***polyamides***.

Nylon is a thermoplastic ***silky*** material that can be ***melt-processed*** into ***fibres, films or shapes***.




Sewing Threads


Backpack


Raincoat


Umbrella

Properties

Creamy colour, hard, tough, resilient to wear, low coefficient of friction, bearing surfaces, self-lubricating, resistant to extremes of temperature, good chemical resistance, machines well, difficult to join except mechanically.

Applications

Bearings, gear wheels, casings for power tools, curtain rail fittings, combs, clothing, stockings, hinges, filaments for toothbrushes


Sleeping bag


Rope


Seat belt


Hose


Tarpaulin


Dental floss

Next >

Polycarbonate

Polycarbonate *is a* naturally transparent thermoplastic, it's usefulness is in its ability to internally transmit light almost as effectively as glass and to withstand impacts far greater than many other commonly used plastics.

Properties

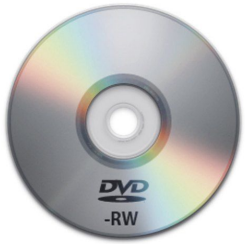
Strong, stiff, hard, tough, transparent engineering thermoplastics that can maintain rigidity up to 140°C and toughness down to -20°C.





Applications

Light diffusers, skylights, riot shields, machine guards, safety spectacles, signs and displays, aircraft interiors and electrical housings.





Next >

Polystyrene (PS)

Polystyrene is a versatile plastic used to make a wide variety of consumer products. There are three different types of polystyrene plastic.





Conventional Polystyrene:

Properties

Light, hard, stiff, colourless, transparent, brittle, low impact strength, safe with food, good water resistance



Applications

Model kits, packaging, disposable plates, cups, utensil's, TV cabinets, containers



Next >

Toughened Polystyrene (HIP):

Properties
Increases impacted resistance, strength, coloured

Applications
Toys, refrigerator linings





Expanded/Foam Polystyrene:

Properties
Buoyant, lightweight, crumbles, good sound/heat insulator

Applications
Sound and heat insulation, packaging







Next >

Polythene (polyethylene)

Polyethylene (Polythene) is one of the world's most popular plastics. It is an enormously versatile polymer.



By altering the formulation and gauge of polyethylene, the producer can adjust the properties of the PE.

The two most common forms of PE are ***LDPE*** and ***HDPE***.



Properties LDPE:

Soft, flexible, translucent, wide range of colours, very tough, weatherproof, good chemical resistance, low water absorption, easily processed by most methods, low cost.

Applications LDPE:

Squeeze bottles, toys, carrier bags, high frequency insulation, chemical tank linings, heavy duty sacks, general packaging.



Next >

Polythene (polyethylene)

Properties HDPE:
Stiffer and harder than LDPE. HDPE is *translucent/waxy, weatherproof, good low temperature toughness (to -60'C), easy to process by most methods, low cost, good chemical resistance*.



Applications HDPE:
Chemical drums, bottles, gas and water pipes, toys, picnic ware, household and kitchenware (buckets, bowls).







Next >

Polypropylene (PP)

Polypropylene is one of the most versatile polymers available with applications, both as a plastic and as a fibre.





Properties

Semi-rigid, translucent, good chemical resistance, tough, good fatigue resistance, integral hinge property, good heat resistance



Applications

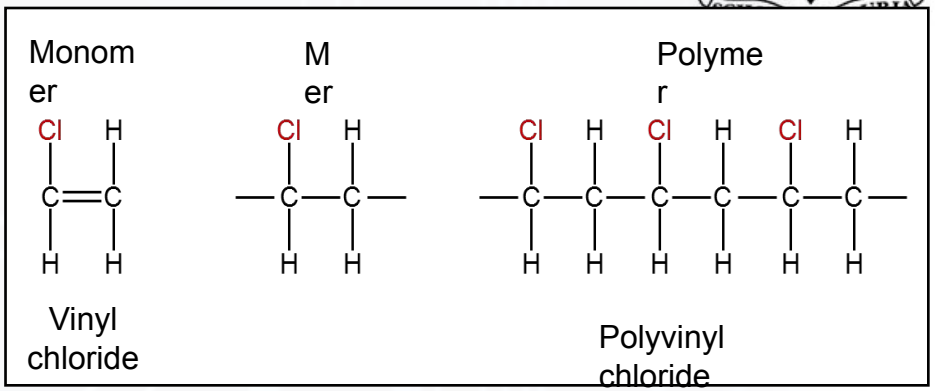
Medical equipment, syringes, containers with integral hinges, string, rope, nets, crates, chair shells, kitchen ware



Next >

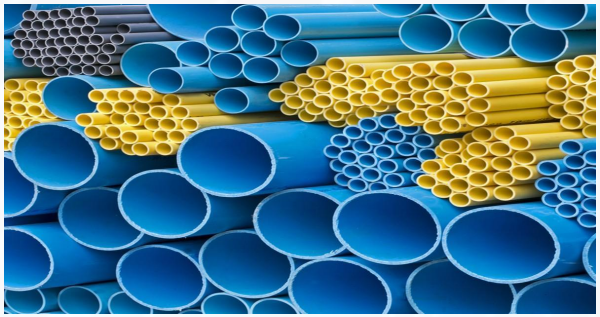
PVC (polyvinyl chloride)

One of the most common thermoplastic polymers is ***PVC (polyvinyl chloride)***.



Properties:
It is ***lightweight***, has ***excellent chemical resistance*** and is ***inexpensive***.

Applications:
PVC is used for *pipes, insulation on electrical cables, construction and even for clothing*.









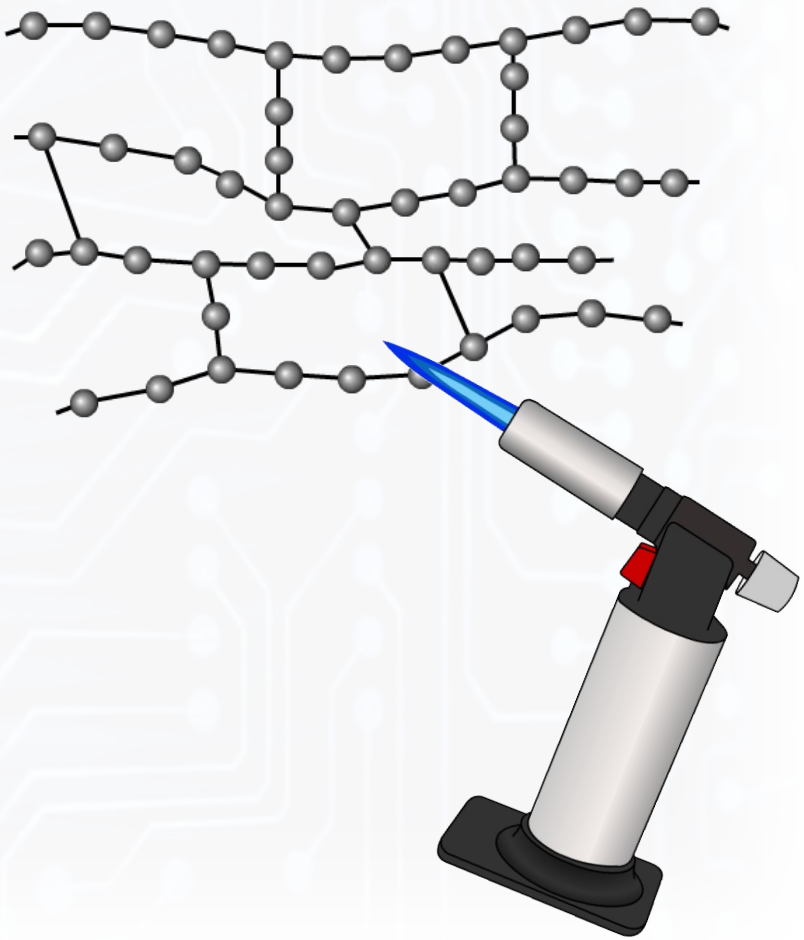
Next >

Thermosetting Polymers

Thermosetting Polymers

Thermosetting polymers consist of long chain molecules cross linked into a network type structure.

During processing a chemical reaction occurs, ***forming the cross linking bonds***.

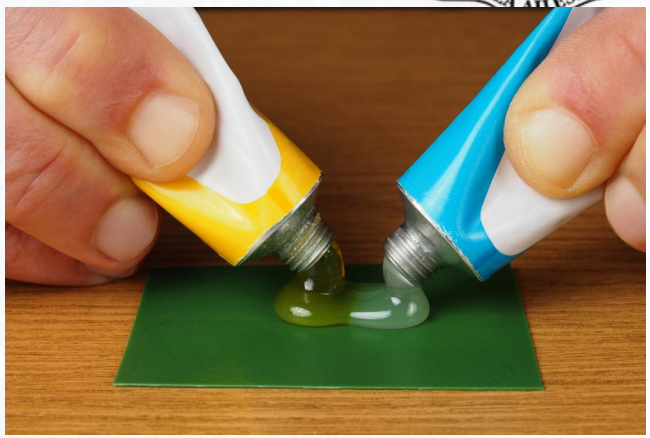


Repeated heating <u>does not soften</u> the material. The mechanical properties are retained unless the polymer starts to burn.

Next >

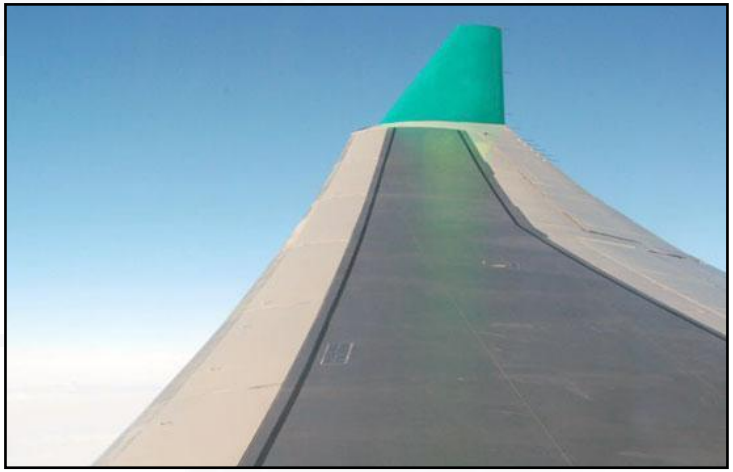
Epoxy Resin

Epoxy is a copolymer formed from the reaction of two chemicals; a ***resin*** and a ***hardener***.



The main uses of ***epoxy*** are as a robust coating and as an engineering adhesive.

Epoxy is also used as the *polymer matrix in fibre reinforced composite materials*, such as those in *some aircraft structures*.



Properties

High strength when re-inforced, good chemical and wear resistance, resists heat to 250ºC, electrical insulator

Applications

Surface coatings, encapsulation of electronic components, adhesives, laminating paper, PCB's

Next >

Polyester Resin

Polyester Resin (more loosely referred to as ‘laminating resin’ or ‘fibreglass resin’) is a polymer. Simple to use and economical, it is a major component in many industries, from construction to aerospace. Offering incredible physical and chemical properties, it’s also comparatively cheap compared to other resins like epoxies and urethanes, yet it offers similar – if not superior – performance in many cases.



Properties

Good strength and stiffness but brittle
Very good temperature, chemical and electrical resistance
Lower cost than other resins



Applications

GRP products, Costume Jewellery, Paperweights, Encapsulations (of Flowers, Coins or Biological Specimens)

Next >

Urea-Formaldehyde

Urea-formaldehyde, also known as ***urea-methanol*** is a non-transparent thermosetting resin or polymer. It is produced from ***urea*** and ***formaldehyde***.

Properties

Stiff, hard, strong, brittle, heat resistant, good electrical insulator, wide range of light colours



Applications

(White) electrical fittings, domestic appliance parts (e.g. knobs), adhesives (wood), coating paper, textiles



Next >

Melamine-Formaldehyde

Melamine-formaldehyde is a hard, thermosetting plastic material made from ***melamine*** and ***formaldehyde***.



Properties

Stiff, hard, strong, scratch resistant, low water absorption, ordourless, stain resistant, resists some chemicals, wide range of colours

Applications

Tableware, decorative laminates for work surfaces, electrical insulation, button



Next >

Polyurethanes



Polyurethane (PU) is a thermoset polymer composed of urethane links. Strong and mouldable, PUs are an extremely versatile polymer used in many applications.



Properties

Hard with high strength. Flexible, tough and low thermal conductivity.

Applications

Foam insulation panels, hoses, surface coatings (paints / varnish) and sealants.



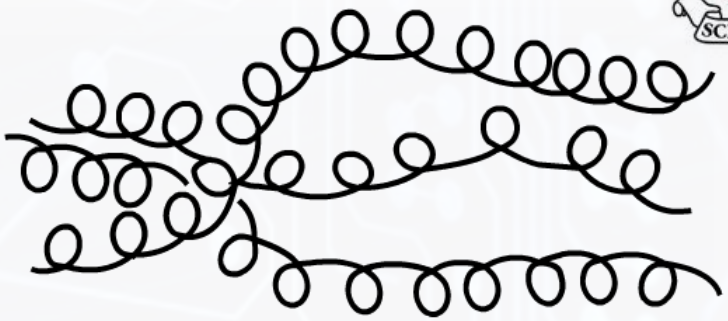


Next >

Elastomers

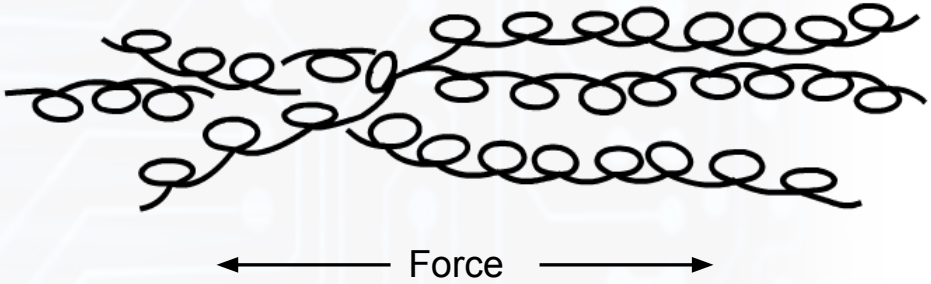
Elastomers

Elastomers are a group of natural and synthetic polymers.



Their distinguishing characteristic is the large amount of elastic deformation when a force is applied.

The structure consists of long chains, heavily cross linked and coiled. When a force is applied, the chains uncoil.



Next >

Elastomers

Rubber is a natural elastomer derived from latex extracted from the rubber tree.

Rubber is ***flexible, elastic and waterproof***.

Natural rubber bands



Robert Benner

Waterproof rubber boots



Rubber can be modified before use by heating it with ***sulphur*** to improve its ***elasticity*** and ***wear resistance (vulcanised rubber)***.

Vulcanised rubber tyre



Kevin Collins

Next >

Elastomers

Synthetic rubbers are a range of polymers derived from oil-based hydrocarbons.

Synthetic rubber can be designed to have a wide range of properties.

Examples of well known synthetic rubbers include *neoprene, nitrile and silicone rubber*.



Neoprene diving suit



Disposable nitrile gloves



Silicone rubber keypad

Next >

Common Thermoplastics – *Properties and Uses*

Thermoplastics		
Material	**Properties and working characteristics**	**Uses**
Polythene (polyethylene) (LDPE)	Low density: tough, common plastic, good chemical resistance, flexible, soft, attracts dust, electrical insulator, wide range of colours	Detergent, squeezy bottles, toys, packaging film, carrier bags, TV cable
(HDPE)	High density: stiffer, harder, high softening point, can be sterilised, waxy feel	Milk crates, bottles, pipes, houseware, buckets, bowls
Polypropylene (PP)	Light, hard, impact resistant even at low temperatures, good chemical resistance, can be sterilised, easily joined, welded, good resistance to work-fatigue, bending, hinges, good mechanically	Medical equipment, syringes, containers with integral hinges, string, rope, nets, crates, chair shells, kitchenware, film
Polystyrene (PS)	a) Conventional: light, hard, stiff, colourless, transparent, brittle, low impact strength, safe with food, good water resistance	Model kits, packaging, disposable plates, cups, utensils, TV cabinets, containers
	b) Toughened: increases impact, strength, pigmented	Toys, refrigerator linings
	c) Expanded/foam: buoyant, lightweight, crumbles, good sound/heat insulator	Sound and heat insulation, packaging
uPVC (Polyvinyl chloride) Plasticised (PVC)	Good chemical, weather resistance, stiff, hard, tough, lightweight, wide colour ranges, needs to be stabilised for outdoor use Soft, flexible, good electrical insulator	Pipes, guttering, bottles, shoe soles, roofing sheets, records, window frames Underseal, hosepipes, wall coverings
Acrylic (Polymethyl methacrylate)	Stiff, hard, clear, very durable, 10X impact resistance of glass, but scratches easily, excellent light transmission, fibre optic qualities, safe with food, good electrical insulator, colours well, easily machined, polishes well	Light units, illuminated signs, record player lids, aircraft canopies, windows, rear car lights/reflectors, furniture, sanitary ware
Nylon (Polyamide)	Creamy colour, hard, tough, resilient to wear, low coefficient of friction, bearing surfaces, self-lubricating, resistant to extremes of temperature, good chemical resistance, machines well, difficult to join except mechanically	Bearings, gear wheels, casings for power tools, curtain rail fittings, combs, clothing, stockings, hinges, filaments for brushes
Cellulose acetate	Tough, hard and stiff (can be made flexible), resilient, light in weight, transparent, non-flammable, easily machined, absorbs some moisture	Pen cases, photographic film, cutlery handles, knobs, lids, spectacle frames, containers
ABS (Acrylonitrile butadienestyrene)	High impact strength and toughness, scratch resistant, light and durable, good appearance, high surface finish, resistant to chemicals	Kitchen ware, cases for consumer durables (e.g. cameras), toys, safety helmets, car components, telephones, food processors/mixers

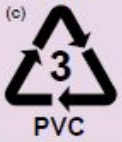
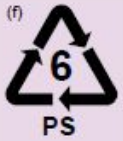
Next >

Common Thermosetting Plastics – *Properties and Uses*

Thermosetting plastics		
Urea-formaldehyde (UF)	Stiff, hard, strong, brittle, heat resistant, good electrical insulator, wide range of light colours, adhesive (Aerolite)	(White) electrical fittings, domestic appliance parts (e.g. knobs), adhesives (wood), coating paper, textile
Melamine-formaldehyde (MF)	Stiff, hard, strong, scratch resistant, low water absorption, odourless, stain resistant, resists some chemicals, wide range of colours	Tableware, decorative laminates for work surfaces, electrical insulation, buttons
Polyester resin (PR)	Stiff, hard, brittle (resilient when laminated GRP), good heat and chemical resistance, electrical insulator, resists ultra-violet light, good outdoors, contracts on curing, takes colour well	Casting, encapsulation, embedding, panels (with GRP), boats, car bodies, chair shells, containers
Epoxy resin (epoxide) (ER)	High strength when re-inforced, good chemical and wear resistance, resists heat to 250°C, electrical insulator, adhesive for bonding unlike materials, low shrinkage	Surface coatings, castings, encapsulation of electronic components, adhesives, laminating paper, PCB, tanks, pressure vessels

Next >

Polymers – *Recycling*

Thermoplastic	ID code	Properties	Applications
PET Polyethylene terephthalate	(a) 	Excellent barrier against atmospheric gases and does not allow gas to escape Does not flavour the food or drink contained in it Sparkling 'crystal clear' appearance Very tough Light – low density	Carbonated (fizzy drinks bottles) Packaging for highly flavoured food Microwaveable food trays
HDPE High density polyethylene	(b) 	Highly resistant to chemicals Good barrier to water Tough and hard wearing Decorative when coloured Light and floats on water Rigid	Unbreakable bottles (for washing-up liquid, detergents, cosmetics, toiletries, etc.) Very thin packaging sheets
PVC Polyvinyl chloride	(c) 	Weather resistant – does not rot Chemical resistant – does not corrode Protects products from moisture and gases while holding in preserving gases Strong, abrasive, resistant and tough Can be made either rigid or flexible	Packaging for toiletries, pharmaceutical products, food and confectionery, water and fruit juices
LDPE Low density polyethylene	(d) 	Good resistance to chemicals Good barrier to water but not to gases Tough and hard wearing Decorative when coloured Very light and floats on water Very flexible	Stretch wrapping (cling film) Milk carton coatings
PP Polypropylene	(e) 	Lightweight Rigid Excellent chemical resistance Versatile – can be made stiffer than polythene or very flexible Low moisture absorption Good impact resistance	Food packaging – yoghurt and margarine pots, sweet and snack wrappers
PS Polystyrene	(f) 	Rigid polystyrene: • Transparent (clear) • Rigid (can be brittle) • Lightweight • Low water absorption Expanded polystyrene (foam): • Excellent impact resistance • Very good heat insulator • Durable • Lightweight • Low water absorption	Food packaging, (e.g. yoghurt pots), CD cases, jewel cases, audio cassette cases, take-away food packaging, egg cartons, fruit, vegetable and meat trays, cups, etc. Packing for electrical and fragile products

Next >

Summary

In this presentation you have:

- explored the structure of polymers
- seen different types of polymers such as thermoplastics, thermosetting polymers and elastomers